Summary of Regression Modeling Concepts

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Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# printr is not loaded here: it inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
# Summarizes key information about statistical objects in tidy tibbles
```

```
library(broom)
# Provides suite of functions to work with regression model
# `broom::tidy()` tibbles
library(broom.helpers)
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")
    )
)

knitr::opts_chunk$set(message = FALSE)
options(digits = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
# use the `filter()` function from dplyr() by default
conflicts_prefer(dplyr::filter)
legend_text_size <- 9
run_graphs <- TRUE
```

1 We use different probability models for different data types

- Binary outcomes: Bernoulli[1] models
- Event rate[2] outcomes: Poisson[3]/Negative binomial[4] models
- Time-to-event outcomes: Survival models
- Catch-all: Gaussian[5] models

2 We use different link functions to connect these models with covariates

- Bernoulli models: logit[6] link
- Count models: log link + offset[7]
- Survival models: log link
- Gaussian models: identity link

Figure 1 shows how the various models we have studied have analogous structures: each row uses model-specific transformations to connect the modeled quantity to a linear predictor[8] $\eta(\underset{\sim}{x}) = \underset{\sim}{x} \cdot \underset{\sim}{\beta} + \eta_0$.

[1] https://morrison-lab.github.io/pds/random-variables.html#def-bernoulli
[2] https://morrison-lab.github.io/pds/distributions.html#def-event-rate
[3] https://morrison-lab.github.io/pds/distributions.html#def-poisson
[4] https://morrison-lab.github.io/pds/distributions.html#def-nb
[5] https://morrison-lab.github.io/pds/random-variables.html#def-normal
[6] binary-outcome-associations.qmd#def-logit-fn
[7] https://morrison-lab.github.io/pds/distributions.html#def-offset
[8] Intro-to-GLMs.qmd#def-linear-predictor

Linear[9] (Gaussian):

$$\underset{\text{mean}}{\underset{\sim}{\mu}} \underset{=}{\overset{=}{\rightleftarrows}} \underbrace{\underset{\sim}{\eta}(\underset{\sim}{x})}_{\substack{\text{linear}\\ \text{predictor}}} = \underset{\sim}{x} \cdot \underset{\sim}{\beta} + \mu(0)$$

Logistic[10] (Binomial):

$$\underset{\text{mean}}{\underset{\sim}{\mu}} \underset{\pi \cdot n}{\overset{\frac{\mu}{n}}{\rightleftarrows}} \underset{\text{probability}}{\underset{\sim}{\pi}} \underset{\frac{\omega}{1+\omega}}{\overset{\frac{\pi}{1-\pi}}{\rightleftarrows}} \underset{\text{odds}}{\underset{\sim}{\omega}} \underset{\exp\{\eta\}}{\overset{\log\{\omega\}}{\rightleftarrows}} \underset{\text{log-odds}}{\underset{\sim}{\eta}} = \underset{\sim}{x} \cdot \underset{\sim}{\beta} + \eta(0)$$

Count (Poisson[11]):

$$\underset{\text{mean}}{\underset{\sim}{\mu}} \underset{\lambda \cdot t}{\overset{\mu/t}{\rightleftarrows}} \underset{\text{rate}}{\underset{\sim}{\lambda}} \underset{\exp\{\eta\}}{\overset{\log\{\lambda\}}{\rightleftarrows}} \underset{\text{log-rate}}{\underset{\sim}{\eta}} = \underset{\sim}{x} \cdot \underset{\sim}{\beta} + \log\{\lambda(0)\}$$

Survival (Cox PH[12]):

$$\underset{\text{mean}}{\underset{\sim}{\mu}} \underset{\int_{t=0}^{\infty} \mathrm{S}(t|\underset{\sim}{x})dt}{\longleftarrow} \underbrace{\mathrm{S}(t|\underset{\sim}{x})}_{\text{survival}} \underset{\exp\{-\Lambda(t|\underset{\sim}{x})\}}{\overset{-\log\{\mathrm{S}(t|\underset{\sim}{x})\}}{\rightleftarrows}} \underbrace{\Lambda(t|\underset{\sim}{x})}_{\text{cumulative hazard}} \underset{\int_0^t \lambda(u|\underset{\sim}{x})\,du}{\overset{\Lambda'(t|\underset{\sim}{x})}{\rightleftarrows}} \underbrace{\lambda(t|\underset{\sim}{x})}_{\text{hazard}} \underset{\exp\{\eta(t|\underset{\sim}{x})\}}{\overset{\log\{\lambda(t|\underset{\sim}{x})\}}{\rightleftarrows}} \underbrace{\eta(t|\underset{\sim}{x})}_{\text{log-hazard}} = \underset{\sim}{x} \cdot \underset{\sim}{\beta} + \eta_0(t)$$

Figure 1: Parallel structures of GLMs[13] and survival models. Each row shows the transformation chain from the modeled quantity to the linear predictor $\eta(\underset{\sim}{x}) = \underset{\sim}{x} \cdot \underset{\sim}{\beta} + \eta_0$, where η_0 denotes the model-specific intercept term, and $\eta_0(t)$ denotes the Cox PH baseline log-hazard term.

3 We use maximum likelihood estimation to fit models to data

- likelihood[14]
- log-likelihood[15]
- score function[16]
- hessian[17]

4 We use asymptotic normality of MLEs to quantify uncertainty about models

- observed information matrix[18]
- expected information matrix[19]
- standard error[20]
- confidence intervals[21]
- p-values[22]

[14] https://morrison-lab.github.io/sds/intro-MLEs.html#def-lik
[15] https://morrison-lab.github.io/sds/intro-MLEs.html#def-loglik
[16] https://morrison-lab.github.io/sds/intro-MLEs.html#def-score
[17] https://morrison-lab.github.io/mds/vector-calculus.html#def-hessian
[18] https://morrison-lab.github.io/sds/intro-MLEs.html#def-oinf
[19] https://morrison-lab.github.io/sds/intro-MLEs.html#def-einf
[20] https://morrison-lab.github.io/sds/estimation.html#def-SE
[21] https://morrison-lab.github.io/sds/inference.html#def-confidence-interval
[22] https://morrison-lab.github.io/sds/inference.html#def-p-value

5 We use (log) likelihood ratios to compare models

Sometimes we adjust these comparisons for model size (AIC[23], BIC[24])

References

[23] Linear-models-overview.qmd#def-aic
[24] Linear-models-overview.qmd#def-bic